WORKSHEET 4.5 : INTERSECTION OF TWO PLANES

2) If $\vec{n_1} \neq k\vec{n_2}$, then 2 planes intersect in a straight line.

a) $\vec{n_1} = (6, 12, -9)$
$\vec{n_2} = (4, -8, 6)$
$\vec{n_1} = -\frac{3}{2}\vec{n_2}$ $\therefore$ parallel $\therefore$ NO

b) $\vec{n_1} = (2, -1, -2)$
$\vec{n_2} = (6, -3, -6)$
$\vec{n_2} = 3\vec{n_1}$ $\therefore$ parallel $\therefore$ NO

c) $\vec{r_1} = (6,0,1) + p(1,1,2) + q(4,2,3)$, $p, q \in \mathbb{R}$
$\vec{r_2} = (1,1,-9) + s(5,3,5) + t(3,1,1)$, $s, t \in \mathbb{R}$

$\vec{n_1} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 2 \\ 3 & 1 & 1 \end{vmatrix}$ $\vec{n_2} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & 3 & 5 \\ 3 & 1 & 1 \end{vmatrix}$

$= (-1, 5, -2)$ $= (-2, 10, -4)$

$\vec{n_2} = 2\vec{n_1}$
$\therefore$ parallel $\therefore$ NO!

d) $\vec{r_1} = (1,1,1) + p\underbrace{(0,0,1)}_{\hat{k}} + q\underbrace{(0,1,0)}_{\hat{j}} \Rightarrow \vec{n_1} = \underbrace{(1,0,0)}_{\hat{i}}$

$\vec{r_2} = (0,0,0) + s\underbrace{(0,0,1)}_{\hat{k}} + t\underbrace{(1,0,0)}_{\hat{i}} \Rightarrow \vec{n_2} = \underbrace{(0,1,0)}_{\hat{j}}$

since $\hat{i} \not\parallel \hat{j}$, $\therefore$ YES!

3a) $x + y - 3z = 4$ ① $\quad \vec{n_1} = (1, 1, -3)$
$x + 2y - z = 1$ ② $\quad \vec{n_2} = (1, 2, -1)$
$\vec{n_1} \not\parallel \vec{n_2}$

② − ①: $-y - 2z = 3$.

Let $z = t$ $\quad -y - 2t = 3$. $\quad x + y - 3z = 4$.

$y = -3 - 2t$ $\quad x = 4 - (-3 - 2t) + 3t$

$= 7 + 5t$

$\therefore \vec{r} = (7, -3, 0) + t(5, -2, 1)$, $t \in \mathbb{R}$

b) $5x-2y+2z=-1$ $\quad \vec{n_1}=(5,-2,2)$ $\quad \vec{n_1}=\vec{n_2}$

$5x-2y+2z=-3$. $\quad \vec{n_2}=(5,-2,2)$ $\quad \therefore$ parallel.

↑ equations not scalar multiples, $\therefore$ parallel & distinct planes

c) $x-3y-z=-3$ ① $\quad \vec{n_1}=(1,-3,-1)$ } $\vec{n_1} \not\parallel \vec{n_2}$

$\ominus$ $2x+4y-z=5$ ② $\quad \vec{n_2}=(2,4,-1)$ }

$-x-7y=-8$

Let $y=t$ $\quad -x-7y=-8$ $\quad x-3y-z=-3$.

$8-7t=x$ $\quad (8-7t)-3t+3=z$.

$11-10t=z$

$\therefore \vec{r}=(8,0,11)+t(-7,1,-10)$, $t\in\mathbb{R}$

d) $x+y+z=1$ ① $\quad \vec{n_1}=(1,1,1)$ } $\vec{n_1} \not\parallel \vec{n_2}$

$\ominus$ $x=0$ ② $\quad \vec{n_2}=(1,0,0)$ }

$y+z=1$

Let $z=t$ $\quad y+z=1$ $\quad x=0$.

$y=1-t$

$\therefore \vec{r}=(0,1,0)+t(0,-1,1)$, $t\in\mathbb{R}$

e) $x+3y-z=4$. ① $\quad \vec{n_1}=(1,3,-1)$ } $\vec{n_2}=2\vec{n_1}$

$2x+6y-2z=8$ ② $\quad \vec{n_2}=(2,6,-2)$ } $\therefore$ parallel.

↑ Equations are scalar multiples ② = ① × 2 $\therefore$ coincident planes

4a) $\left[\begin{array}{ccc|c} 3 & -7 & 1 & 12 \\ 1 & 1 & -2 & -3 \end{array}\right]$

b) $\left[\begin{array}{ccc|c} -4 & 3 & 2 & 4 \\ 0 & 2 & -5 & 5 \end{array}\right]$

c) $\left[\begin{array}{ccc|c} 1 & 0 & 4 & 16 \\ 0 & 1 & -8 & -2 \end{array}\right]$

d) $\left[\begin{array}{ccc|c} 6 & 5 & -2 & 4 \\ -2 & 5 & 3 & -4 \end{array}\right]$

5a) $x+4z=9$
$y-6z=4$

b) $8x-2y+3z=-6$
$2x-6y-6z=9$

c) $5x-10z=8$
$3y-4z=6$

d) $x+4z=0$
$y+9z=0$

6a) $\left[\begin{array}{ccc|c} 1 & 2 & 7 & 4 \\ 1 & 3 & 3 & 1 \end{array}\right] \rightarrow \overset{R1-R2}{\left[\begin{array}{ccc|c} 1 & 2 & 7 & 4 \\ 0 & -1 & 4 & 3 \end{array}\right]} \rightarrow \overset{R1+2R2}{\left[\begin{array}{ccc|c} 1 & 0 & 15 & 10 \\ 0 & 1 & 4 & 3 \end{array}\right]}$

Let $z=t$ $\quad y+4z=3$ $\quad x+15z=10$

$y=3-4t$ $\quad x=10-15t$

$\therefore (x,y,z) = (10-15t, 3-4t, t)$

$= (10,3,0) + t(-15,-4,1)$, $t \in \mathbb{R}$

b) $\left[\begin{array}{ccc|c} 1 & -4 & 3 & 5 \\ 2 & 1 & 6 & 0 \end{array}\right] \rightarrow \overset{2R1-R2}{\left[\begin{array}{ccc|c} 1 & -4 & 3 & 5 \\ 0 & -9 & 0 & 10 \end{array}\right]} \rightarrow \overset{\substack{9R_1-4R_2 \\ -\frac{1}{9}R_2}}{\left[\begin{array}{ccc|c} 9 & 0 & 27 & 5 \\ 0 & 1 & 0 & -\frac{10}{9} \end{array}\right]}$

$\rightarrow \overset{\frac{1}{9}R1}{\left[\begin{array}{ccc|c} 1 & 0 & 3 & \frac{5}{9} \\ 0 & 1 & 0 & -\frac{10}{9} \end{array}\right]}$

Let $z=t$, $\quad y=\frac{-10}{9}$, $\quad x+3z=\frac{5}{9}$

$x=\frac{5}{9}-3t$

$\therefore (x,y,z) = \left(\frac{5}{9}-3t, \frac{-10}{9}, t\right)$

$= \left(\frac{5}{9}, \frac{-10}{9}, 0\right) + t(-3,0,1)$, $t \in \mathbb{R}$

c) $\left[\begin{array}{ccc|c} 2 & 8 & 2 & 7 \\ 1 & 4 & -1 & 3 \end{array}\right] \xrightarrow{R1-2R2} \left[\begin{array}{ccc|c} 2 & 8 & 2 & 7 \\ 0 & 0 & 4 & 1 \end{array}\right] \xrightarrow{2R1-R2} \left[\begin{array}{ccc|c} 4 & 16 & 0 & 13 \\ 0 & 0 & 4 & 1 \end{array}\right]$

$\rightarrow \begin{matrix} \frac{1}{4}R1 \\ \frac{1}{4}R2 \end{matrix} \left[\begin{array}{ccc|c} 1 & 4 & 0 & \frac{13}{4} \\ 0 & 0 & 1 & \frac{1}{4} \end{array}\right]$

$\therefore z = \frac{1}{4}$ Let $y = t$, $x + 4y = \frac{13}{4}$

$x = \frac{13}{4} - 4t$

$\therefore (x, y, z) = \left(\frac{13}{4} - 4t, t, \frac{1}{4}\right)$

$= \left(\frac{13}{4}, 0, \frac{1}{4}\right) + t(-4, 1, 0), \ t \in \mathbb{R}$

d) $\left[\begin{array}{ccc|c} 4 & -8 & -3 & 6 \\ -3 & 6 & 1 & -2 \end{array}\right] \xrightarrow{3R1+4R2} \left[\begin{array}{ccc|c} 4 & -8 & -3 & 6 \\ 0 & 0 & -5 & 10 \end{array}\right] \xrightarrow{-\frac{1}{5}R2} \left[\begin{array}{ccc|c} 4 & -8 & -3 & 6 \\ 0 & 0 & 1 & -2 \end{array}\right]$

$\rightarrow \xrightarrow{R1+3R2} \left[\begin{array}{ccc|c} 4 & -8 & 0 & 0 \\ 0 & 0 & 1 & -2 \end{array}\right] \rightarrow \xrightarrow{\frac{1}{4}R1} \left[\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 0 & 1 & -2 \end{array}\right]$

$z = -2$, Let $y = t$ $x - 2y = 0$

$x = 2t$

$\therefore (x, y, z) = (2t, t, -2)$

$= (0, 0, -2) + t(2, 1, 0), \ t \in \mathbb{R}$

e) $\left[\begin{array}{ccc|c} 3 & 2 & -6 & 5 \\ 2 & 3 & -9 & -10 \end{array}\right] \xrightarrow{2R1-3R2} \left[\begin{array}{ccc|c} 3 & 2 & -6 & 5 \\ 0 & -5 & 15 & 40 \end{array}\right] \xrightarrow{-\frac{1}{5}R2} \left[\begin{array}{ccc|c} 3 & 2 & -6 & 5 \\ 0 & 1 & -3 & -8 \end{array}\right]$

$\rightarrow \xrightarrow{R1-2R2} \left[\begin{array}{ccc|c} 3 & 0 & 0 & 21 \\ 0 & 1 & -3 & -8 \end{array}\right] \rightarrow \xrightarrow{\frac{1}{3}R1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 7 \\ 0 & 1 & -3 & -8 \end{array}\right]$

Let $z = t$, $y - 3z = -8$, $x = 7$

$y = -8 + 3t$

$\therefore (x, y, z) = (7, -8 + 3t, t)$

$= (7, -8, 0) + t(0, 3, 1), \ t \in \mathbb{R}$

f) $\left[\begin{array}{ccc|c} 6 & 8 & -3 & 9 \\ 10 & -2 & -5 & 15 \end{array}\right] \xrightarrow{5R1-3R2} \left[\begin{array}{ccc|c} 6 & 8 & -3 & 9 \\ 0 & 46 & 0 & 0 \end{array}\right] \xrightarrow{\frac{1}{46}R2} \left[\begin{array}{ccc|c} 6 & 8 & -3 & 9 \\ 0 & 1 & 0 & 0 \end{array}\right]$

$\rightarrow \xrightarrow{R1-8R2} \left[\begin{array}{ccc|c} 6 & 0 & -3 & 9 \\ 0 & 1 & 0 & 0 \end{array}\right] \rightarrow \xrightarrow{\frac{1}{6}R1} \left[\begin{array}{ccc|c} 1 & 0 & -\frac{1}{2} & \frac{3}{2} \\ 0 & 1 & 0 & 0 \end{array}\right]$

$\therefore y = 0$, let $z = t$, $x - \frac{1}{2}z = \frac{3}{2}$

$x = \frac{3}{2} + \frac{1}{2}t$

$\therefore (x, y, z) = \left(\frac{3}{2}, 0, 0\right) + t\left(\frac{1}{2}, 0, 1\right), \ t \in \mathbb{R}$